

Yernar Mars

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Education

Nazarbayev Intellectual School of Astana (NISA)

Sep 2021 – May 2027

IB Diploma Programme

Astana, Kazakhstan

- HL: Mathematics (Analysis & Approaches), Digital Society, English B. SL: Physics, Computer Science, Russian A.
- **“Orken” Presidential Merit Scholarship** (government): full 6-year funding (~\$25,000); ranked 4th of 2,000+ nationally (top 0.2%).
- **IB Personal Project** — ranked 1st of 178 students; qualified for the International IBSA Conference (top personal projects, 7 IB schools worldwide).
- SAT: [SAT score] | IELTS: [IELTS score]

Honors & Awards

International & government-issued (Ministry of Education / DARYN / international brands)

- **Infomatrix Asia** (Ministry of Education), International — Bronze 2026 (6th of 3,000, **top 0.2%**, 13 countries; *Avatar Robot*) & 2025 (12th of 30, *KokysGet*).
- **World Robot Olympiad (WRO) 2026** — 2nd place (Astana; *Avatar Robot*); qualified for the National stage.
- **FIRST Tech Challenge (FTC) 2026** — Motivate, Reach, and Winning Alliance Captain awards; advanced to the Central Asia stage (21st of 40).
- **National Fibonacci Robot Olympiad** (Ministry of Education RK) — 3rd of 63, 2026 (*Avatar Robot*); Judges' Choice & 4th of 37, 2025, qualified for international finals in Antalya, Turkey.
- **National RSPC “Daryn”** (Ministry of Education) — 3rd place, Robotics & Intelligent Machines track (*SoilyGet*).
- **National “Bilim Shyny”** (DARYN, government), 2026 — Silver, 2nd of 300, “Technology of the Future.”

Startup & hackathon wins — 610,000+ KZT in cash prizes

- **1st place** — BECO National Hackathon 2025 (1/30; 150,000 KZT); Synergy Hackathon 2026 (100,000 KZT); Young Visionaries Shark Tank 2025 (1/21; 60,000 KZT); BrightFuture National Hackathon 2024 (1/20; 50,000 KZT).
- **3rd place** — Caspian Startup Battle 2026 (AOO NIS; 3/300; 100,000 KZT); International Startup Cup 2026 (3/320, 4 countries; *Avatar Robot*); National NIS Hack 2026 (AOO NIS; 3/110).
- **2nd place** — IT FEST Startup Battle 2025 (2/20; 100,000 KZT; *Avatar Robot*); Oylab Startup Battle (2/113; 50,000 KZT).

Selective programs (2026): Pioneer Academics Research · LaunchX Entrepreneurship · NYU Pre-College.

Research

Minimum-dose, contrast-free pulmonary-hypertension screening

Aug 2024 – Present

Lead student researcher — mentored by 7 Nazarbayev University professors (Kazakhstan's #1 university) and a UMC Heart Center cardiologist

Astana, KZ

- Building an AI pipeline that measures pulmonary-artery and aorta diameters on **low-dose, non-contrast chest CT** and computes the PA:aorta ratio (a validated marker for pulmonary hypertension) — to find the minimum radiation dose at which screening stays reliable, avoiding both contrast dye and unnecessary radiation.
- Aorta segmentation held **12/12 integrity across a 2.6× dose range** (7.84–20.75 mGy) and 3 reconstruction kernels; ascending-aorta diameter measured to ± 1.7 –2.8 mm within-scan precision (~ 5 –7% CV) on a 3,673-file DICOM dataset.
- Built the PA:aorta ratio + PH-flag pipeline (PA model license-gated, pending). To be validated in radiologists' labs toward clinical use; selected to present at the national *Zhas & Healthy* symposium.

Engineering Projects

SoilyGet — Autonomous soil-monitoring robot

Mar 2024 – Present

Lead hardware & systems engineer

Astana, KZ

- Built a **full-sized**, four-legged wheeled robot that autonomously augers soil samples to a **25 cm** depth, with a servo-driven **4-joint** manipulator modeled in **Fusion 360** and custom **3D-printed** parts. pXRF + field sensors feed an AI contamination map; first-authored the research paper and placed **3rd** at the National RSPC “Daryn” competition (Robotics & Intelligent Machines).

KokysGet — Vision-based waste sorter

Jan 2024 – Present

Founder — hardware, ML & deployment

Astana, KZ

- Trained a custom **YOLOv8** model (~800 self-labeled images) deployed end-to-end on an **ESP32-C3** + Raspberry Pi Camera feeding a **Raspberry Pi**, with servo sorting and a **3D-printed (STL)** chassis. Won BECO Hackathon 2025; Bronze, Infomatrix Asia 2025.

Avatar Robot — Teleoperation system for inclusive employment

Jan 2025 – Present

Founder

Astana, KZ

- Built a robot avatar that mirrors an operator's hand gestures in real time (**MediaPipe** hand-tracking → servo manipulator), with a Java-controlled drivetrain and web teleoperation interface, letting people with disabilities remotely staff a café. Piloted at Peek-a-Boo café (Astana) with an initial client. Bronze, Infomatrix Asia 2026 (6th of 3,000, top 0.2%); 2nd, WRO 2026; 3rd, National Fibonacci 2026.

HeartCareAI — Structural heart-defect detection

Aug 2024 – Present

Founder

Astana, KZ

- Built a **YOLOv8**-based classifier that flags structural heart defects (septal “holes”) from chest X-rays, deployed as a Flask diagnostic web tool. [FILL: a real metric, e.g. accuracy/dataset size, if you have one]

Leadership & Experience

Mentoria — Peer-to-peer STEM academy & national hackathon

Dec 2025 – Present

Co-founder & Director

Astana, KZ (Remote)

- Founded a peer-to-peer STEM academy; grew an opportunities channel to 1,195 subscribers and peer-teaching groups (294 + 40 members) across math, physics, chemistry, biology, programming, SAT, and IELTS.
- Organized a free national student hackathon — 330 teams registered — hosted at Nazarbayev University, with a 1,000,000 KZT prize pool and a Bahandi developer internship for the winning team, funded by Bahandi (a major Kazakhstani fast-food chain).

AI automation (freelance)

2023 – Present

Founder & developer

Remote

- Build and sell LLM-based reply and booking assistants for clinics (website chat + Instagram/Facebook DMs); revenue helps cover family expenses and self-funds robotics prototypes.

QApps — EdTech startup: centralized university-admissions platform (\$2M valuation)

Summer 2026

Software engineering intern

Astana, KZ

- Selected as 1 of 3 interns from 30 nationwide applicants; build and debug features in Next.js/Nest.js, test in Docker, and ship via pull requests merged by a mentor. Contributed during growth to 4,000 users and \$50K ARR.

FIRST Tech Challenge — Team Spirit

Oct 2025 – May 2026

Lead programmer (Java)

Astana, KZ

- Programmed the competition robot (Java); **5–0** league record, **5–3** at the Central Asia stage (21st of 40).

Shenber — Summer STEM Camp

Summer 2026

Co-founder & instructor

Astana, KZ

- Co-founded *Shenber*, a paid summer STEM camp: taught 33 children (60,000 KZT each; ~2,000,000 KZT revenue), ~5 hours/day; led curriculum and instruction.

Robotics instructor & Jasyl El community service

2023 – 2026

Teacher & volunteer

Astana, KZ

- Teach robotics to children ages 8–14 — 20 students across 3 cohorts (80,000 KZT/month); completed 250 hours of paid environmental field work with Jasyl El (2 months, ~30 hrs/week, 60,000 KZT/month).

Skills

Programming: Python, Java, C/C++ (Arduino), JavaScript (Next.js / Nest.js). **ML / AI:** PyTorch, YOLOv8, mU-Net / TotalSegmentator, CNNs, computer vision, LLM application building. **Tools:** Docker, Git, Flask, ESP32 / Raspberry Pi, 3D printing, Onshape & Fusion 360 (15+ CAD models). **Languages:** Kazakh (native), Russian (fluent), English (fluent, IB English B HL).